Cap and Trade with Imperfect Hedging

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Cap and Trade

Emissions externalities + no other friction $\Rightarrow$ Cap and trade with emissions caps reflecting the SCC achieves first best

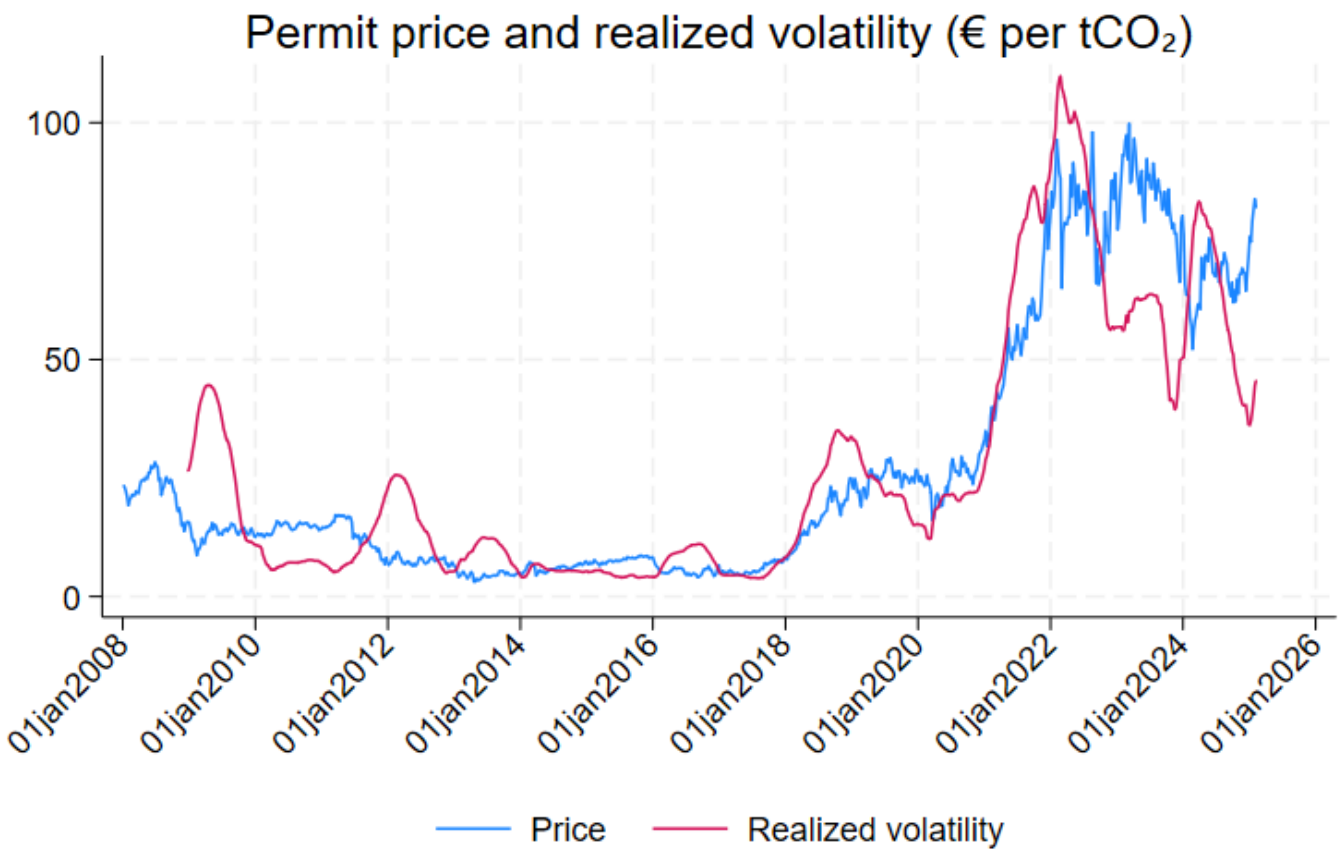
Cap-and-trade systems cover 23% of global emissions in 2025

SCC is highly uncertain due uncertainty about model structure, parameters, and preferences

> *"The range of estimates is large (...) In the past 10 years, estimates of the Social Cost of Carbon have increased from $9 per ton CO2 to $40 for a high discount rate and from $122 per ton CO2 to $525 for a low discount rate."* (Tol, 2023)

$\Rightarrow$ Optimal cap is uncertain $\Rightarrow$ Emitters face transition risk

Example: EU ETS Permit Price



Volatility of compliance cost $\simeq$ 40% of emitters' pre-tax profits

Hedging Under Financial Constraints

Price risk creates demand for hedging.

Draghi Report (2024):

> *"The hedging component (i.e., removing carbon price uncertainty) could also be fulfilled by sufficient ex ante purchases of ETS allowances as the later are 'bankable' (i.e., unused allowances can be saved for later use). Frontloading purchases of ETS allowances would, however, require up-front financing and may hit the financing constraints of companies."*

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This Paper:

1. How do financial frictions affect hedging/pricing of permits?
   - Evidence of hedging/pricing distortions from the EU ETS
   - Model with financial friction rationalizing these facts
2. How do financial frictions affect the optimal policy?
   - Determine second best and its implementation

Literature

- Credit rationing and carbon policy (Rola-Janicka Doettling 2025, Heider Inderst 2022, Oehmke Opp 2025)

  Our paper: Imperfect hedging and carbon policy

- ETS price risk and investment (Kaenzig 2023, Fuchs Stroebel Terstegge 2024)

  Our paper: ETS price risk and hedging

- Exposure to/pricing of transition risk (Bolton Kacperczyk 2021, Sautner et al 2023, etc.)

  Our paper: How frictions affect hedging and pricing of transition risk

Road Map

1. Stylized facts from the EU ETS

2. Model

1. Stylized Facts from the EU ETS

EU ETS

Started in 2008 (after experimental phase 2005-2007)

Firms must surrender permits to cover their yearly GHG emissions

$\simeq$ 50% of permits auctioned, 50% freely allocated

Secondary trading: spot (one-day ahead futures), futures (most liquid is next December), and options

Market participants: emitters and financial firms

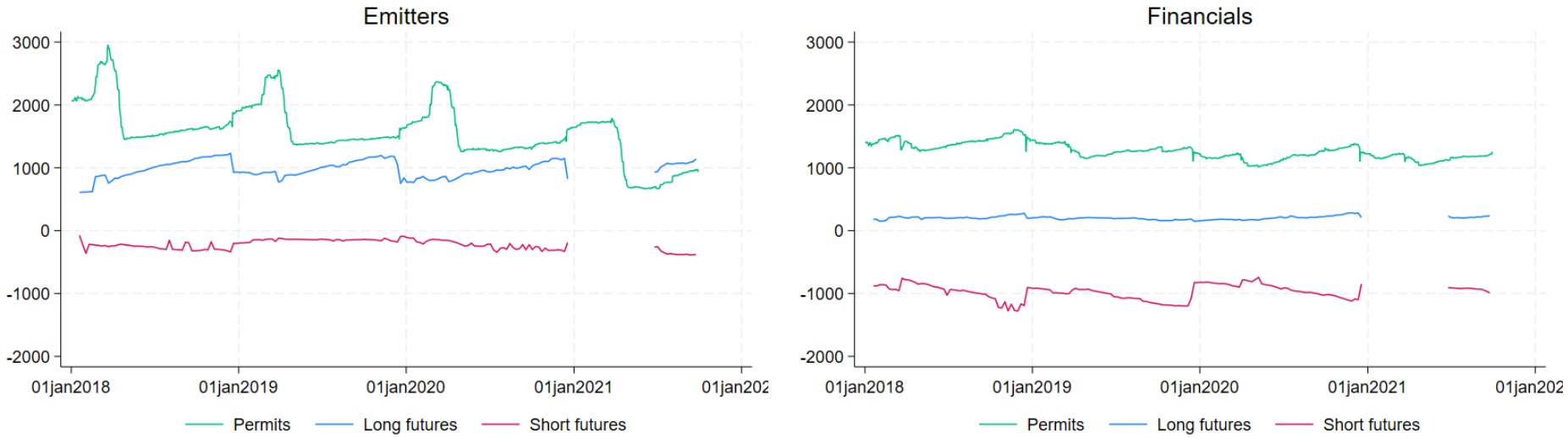
Key Facts

Fact 1 (Risk): Permit price volatility is high and represents 40% of emitters' pre-tax profits

Fact 2 (Hedging):

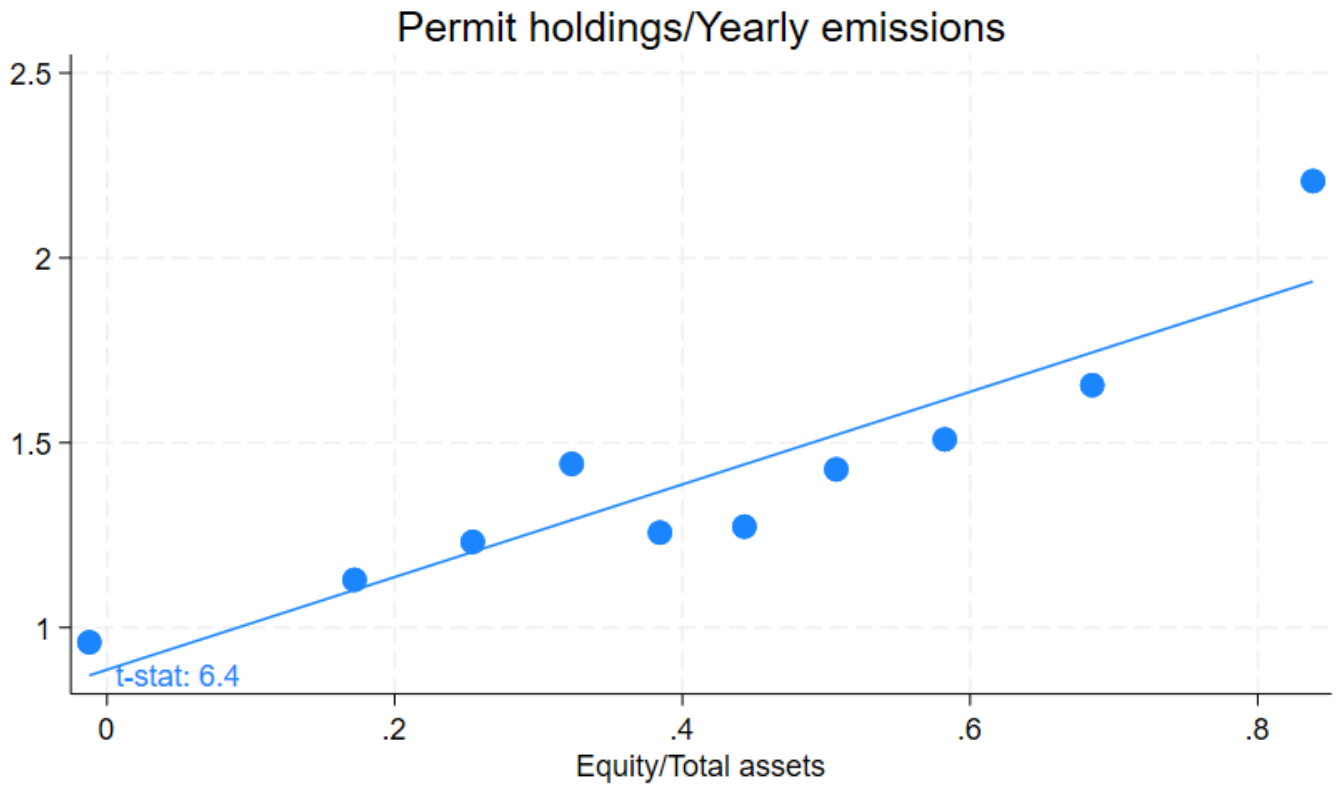
- Emitters buy spot and futures
- Financials sell futures and hedge by buying spot

Holdings of permits and futures (million tCO_2)



Key Facts

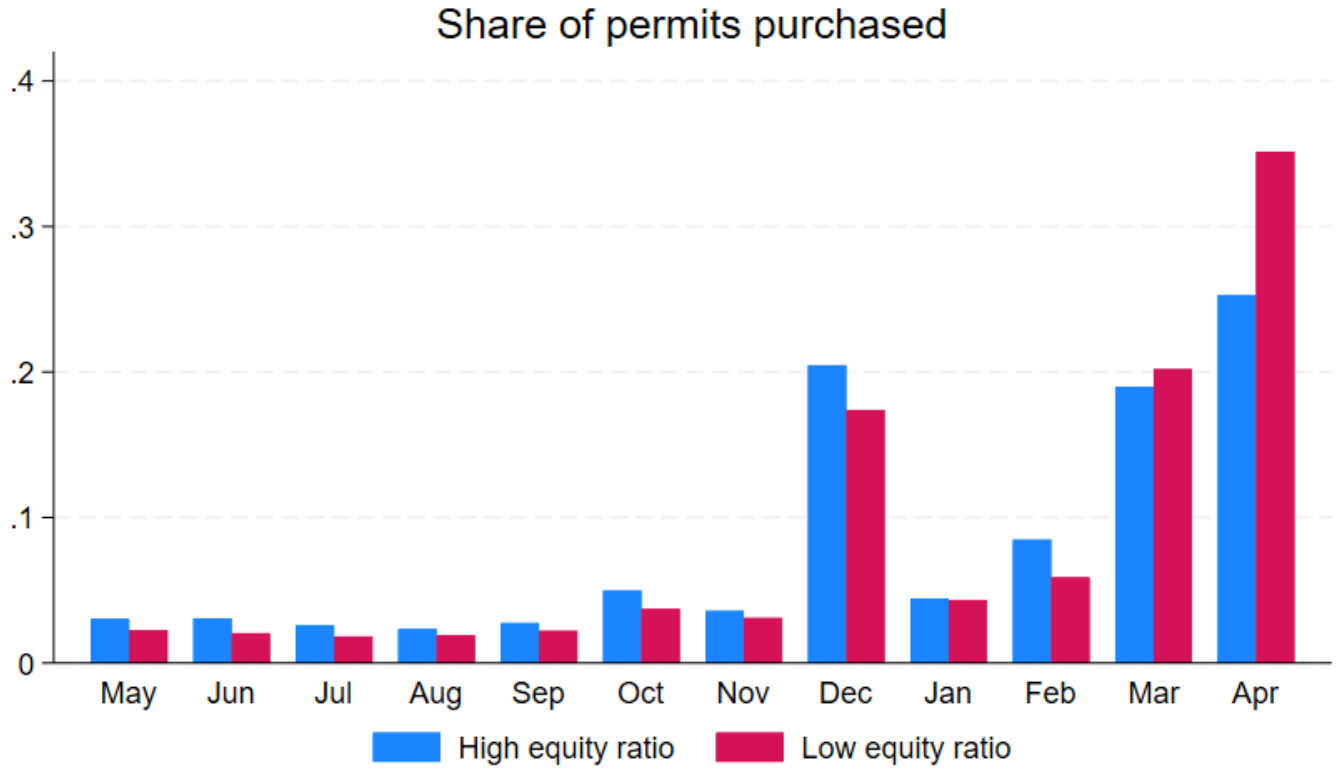
Fact 3a (Financial Constraints and Hedging):
Less capitalized emitters hold less permits



Interpretation: Buying permits requires upfront financing ⇒ costly for firms with binding financial constraint

Key Facts

Fact 3b (Financial Constraints and Hedging):
Less capitalized emitters delay permit purchases



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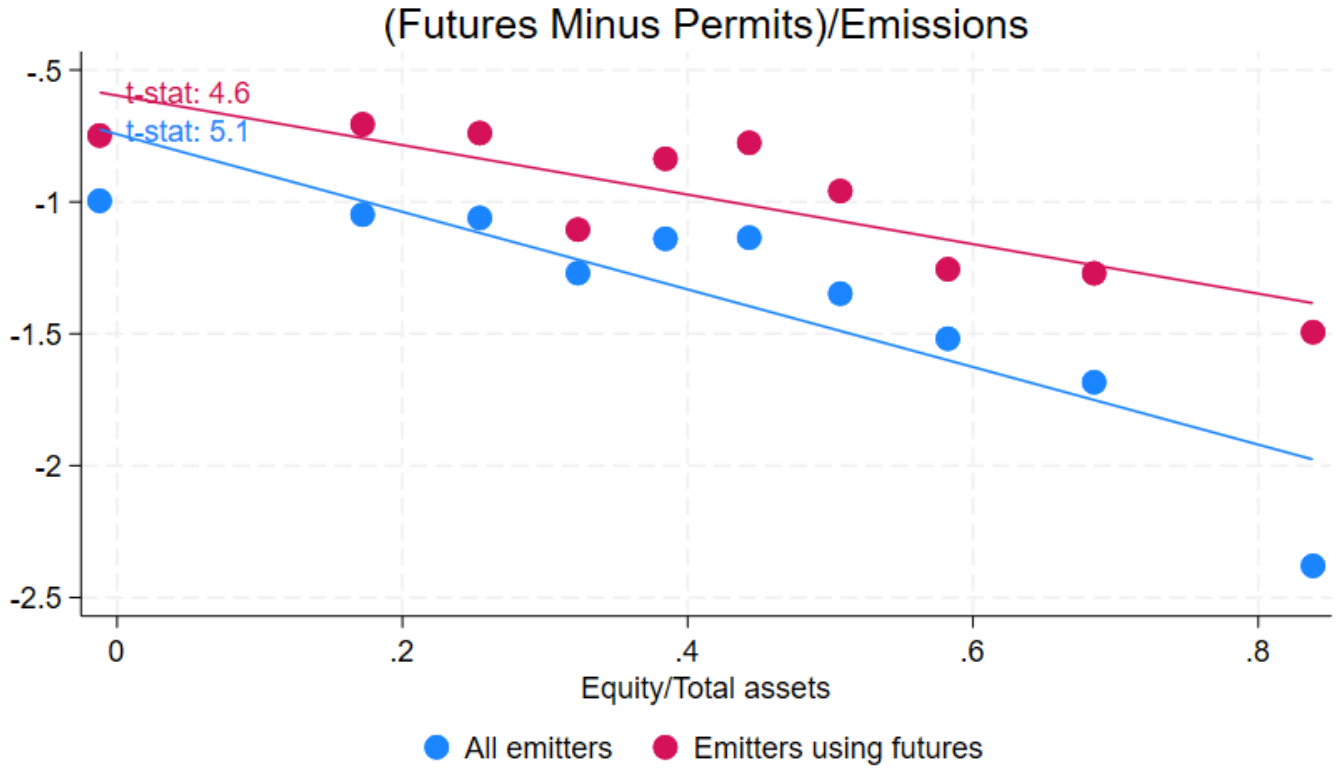
Key Facts

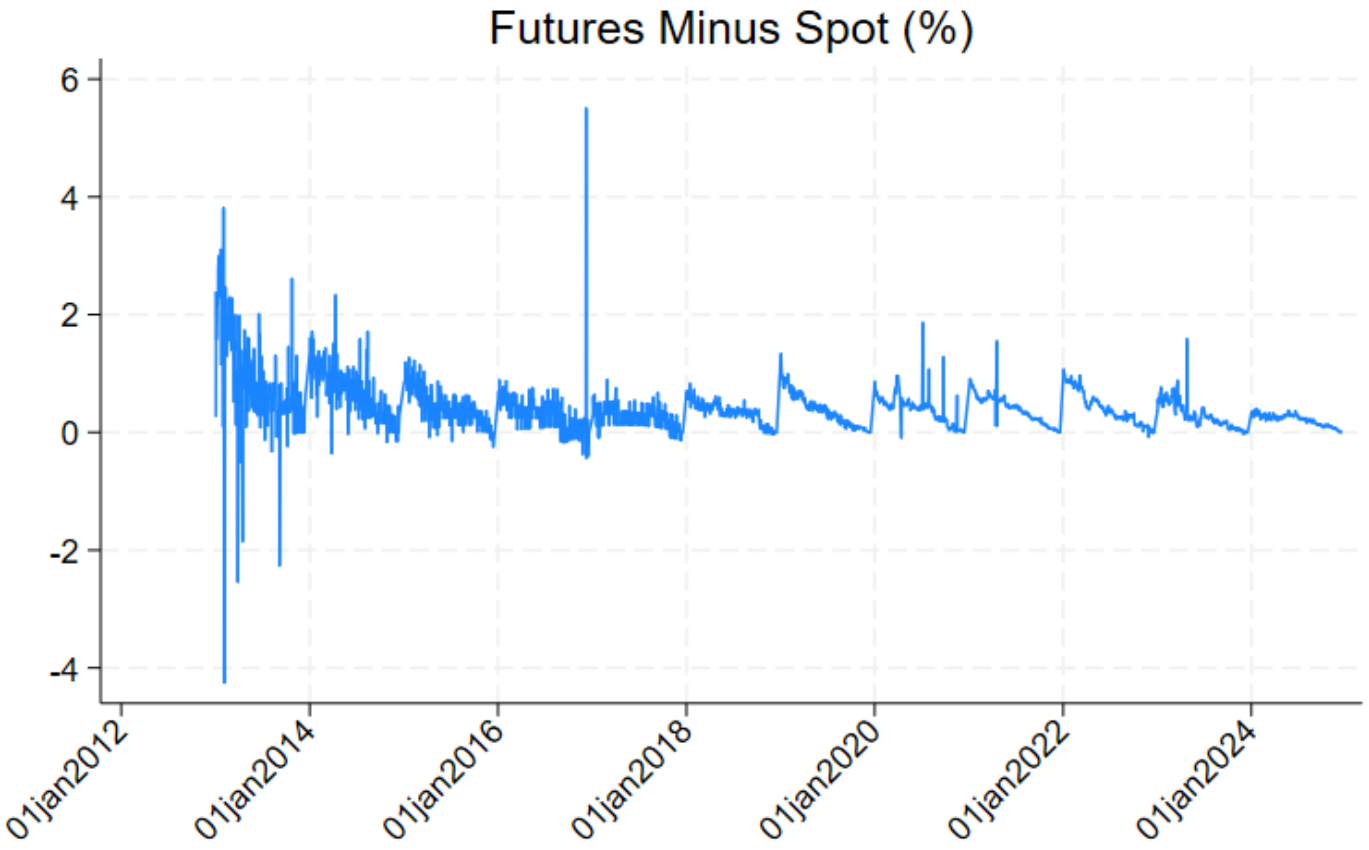
Fact 3c (Financial Constraints and Hedging):
Less capitalized emitters buy futures rather than spot



Interpretation: Buying spot requires upfront financing

Key Facts

Fact 4 (Basis): Permit futures trade at a discount relative to spot (basis $\simeq$ 80 bps per year)



Interpretation: Financial constraints $\Rightarrow$ permits costlier than future

2. Model

Model

Two dates $t = 0, 1$. State realized at $t = 1$: high cap ($s = h$) or low cap ($s = \ell$)

Two types of agents: Emitters and Financials

At $t = 0$:

Government issues $\overline{n}_0$ permits and rebates proceeds to emitters

Emitters and financials trade permits and Arrow securities for each state

Emitters' budget constraint

$$\underbrace{p_0 n_E}_{\substack{\text{permits}\\ \text{holdings}}} + \underbrace{\sum_{s=\ell,h} q_s a_{Es}}_{\substack{\text{Arrow sec.}\\ \text{holdings}}} \leq \underbrace{p_0 \overline{n}_0}_{\text{rebate}}$$

Model

At $t = 1$ in state $s \in \{h, \ell\}$:

Government issues $\overline{n}_s$ permits, $\overline{n}_\ell < \overline{n}_h$

Emitters choose emissions e_s to produce $f(e_s)$, $f' > 0$, $f'' < 0$

Financials endowed with y

E and F consume, utility $u(c_{is}) - \delta_s \overline{e}_s$, social cost of agg. emissions $\delta_s \overline{e}_s$

Emitters' budget:

$$\underbrace{c_{Es}}_{\text{cons.}} \leq \underbrace{f(e_s)}_{\text{prod.}} - \underbrace{p_s e_s}_{\text{cost of permits}} + \underbrace{p_s \overline{n}_s}_{\text{rebate}} + \underbrace{p_s n_E}_{\text{permits held}} + \underbrace{a_{Es}}_{\text{Arrow held}}$$

Despite full rebate, emitters face risk of low production (due to low cap)

$\Rightarrow$ In equilibrium emitters purchase insurance against low cap from financials: Arrow securities in the model, derivatives in practice $\rightarrow$ **Fact 2**

Incentive Constraint

- **Incentive constraint:** Agents can abscond with fraction θ of their time 1 resources

$$c_{Es} \geq \theta\left[f(e_s) + p_s n_E\right] \qquad c_{Fs} \geq \theta\left[y + p_s n_F\right]$$

$\Rightarrow$ Limits how much insurance can be provided

State ℓ: emitters produce little, financials transfer resources to emitters, but not as much as in the first best when financials' IC binds

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  - Permit price $<$ price of replicating portfolio of Arrows $\Rightarrow$ Basis $\rightarrow$ **Fact 4**

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- Extension: 2 types of emitters with different endowment at time 0
  - Emitters with lower endowment $\Rightarrow$ higher liabilities $\Rightarrow$ tighter IC $\Rightarrow$ buy less permits at time 0 (buy permits later) $\rightarrow$ **Facts 3a, 3b, 3c**

Optimal Policy

- Planner chooses consumption and emissions to maximize

$$\sum_{i=E,F} \alpha_i E\left[u(c_{is}) - \delta_s e_s\right]$$

subject to resource constraint $c_{Es} + c_{Fs} \leq f(e_s) + y$

and incentive constraints $\begin{array}{l} c_{Es} \geq \theta\, f(e_s) \\ c_{Fs} \geq \theta\, y \end{array}$

- SCC is higher in state ℓ than in state h: $\delta_\ell > \delta_h$

- Compare the planner's solution without IC (first best) and with IC (constrained optimum)

Optimal Policy

Result 1: The constrained-optimal emissions cap is **less responsive** to the SCC than in the first best

- Binding ICs limit risk sharing $\Rightarrow$ planner reduces transition risk

Result 2: The constrained-optimal emissions cap depends on agents' **Pareto weights** (e.g., political influence), whereas it doesn't in the first best

- First best: emissions only pinned down by SCC (while Pareto weights pin down transfers)
- Constrained optimum: transfers are constrained $\Rightarrow$ transfer utility by distorting emissions

Implementation

Result 3: Constrained optimum can be implemented with **cap and trade** and transfers

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Result 3: Constrained optimum can be implemented with **cap and trade** and transfers

Result 4: Requires zero permits issuance at $t = 0$

- Because storage of permits from $t = 0$ to $t = 1$ tightens IC
- **No front-loading** of permits issuance is consistent with EU's policy

Wrap-Up

- Cap and trade $\Rightarrow$ transition risk (due to uncertainty in SCC) $\Rightarrow$ demand for hedging

- Financial constraints $\Rightarrow$ limited hedging $\Rightarrow$ distorts optimal policy
  - Cap-and-trade remains optimal with zero front-loading of permits issuance
  - Optimal cap is less responsive to SCC
  - Optimal cap depends on political influence